

In Exercises 23 and 24, key statements from this section are either quoted directly, restated slightly (but still true), or altered in some way that makes them false in some cases. Mark each statement True or False, and justify your answer. (If true, give the approximate location where a similar statement appears, or refer to a definition or theorem. If false, give the location of a statement that has been quoted or used incorrectly, or cite an example that shows the statement is not true in all cases.)

- 23) a) Every elementary row operation is reversible.
Ans: This statement is True. It is given in Page 6, last paragraph.
- b) A 5×6 matrix has six rows.
Ans: On Page 4, in the last paragraph of the section Matrix Notation, it is given that an $m \times n$ matrix has m rows and n columns.
∴ A 5×6 matrix has five rows. So, the statement is False.
- c) The solution set of a linear system involving variables $x_1, \dots, x_n$ is a list of numbers $(s_1, \dots, s_n)$ that makes each equation in the system a true statement when the values $s_1, \dots, s_n$ are substituted for $x_1, \dots, x_n$, respectively. It is a generalization of Example 1 in Page 6, where the solution set is verified by substituting into the variables.
Ans: This statement is True. ~~It~~ It is a generalization of Example 1 in Page 6, where the solution set is verified by substituting into the variables.
- d) Two fundamental questions about a linear system involve existence and uniqueness.
Ans: This statement is True. We conclude this from the white box given under the section Existence and Uniqueness Questions on Page 7.
- 24) a) Elementary row operations on an augmented matrix never change the solution set of the associated linear system.
Ans: This statement is True. We conclude this from the second paragraph of Page 7.